

Wizardry Under Uncertainty: Monte Carlo Methods for Bidding in the Card Game Wizard

Joana Mizrahi and Jonathan Dumanski

Stanford University
{jmizrahi, jdumansk}@stanford.edu

Abstract

We develop an agent for Wizard, a trick-taking card game where players bid on how many tricks they will win, then play cards to achieve their bid. The game presents a challenging POMDP: opponent hands are hidden, and the asymmetric scoring function heavily penalizes inaccurate bids. We use Monte Carlo Tree Search (MCTS) with state sampling for card play and forward search with rollouts for bid evaluation. Our experiments reveal that online planning for bidding is far more important than card play optimization: an agent with online bidding achieves 95% win rate (+237 points improvement), while MCTS card play alone provides only +88 points. We also compare particle filtering (weighting sampled opponent hands by bid consistency) against uniform sampling, finding that uniform sampling slightly outperforms particle filtering (54% vs 46% win rate). These results suggest that in Wizard, and perhaps other trick-taking games with bidding, the commitment decision (what to bid) may matter more than the tactical execution (how to play cards).

Introduction

Wizard is a trick-taking card game that combines strategic bidding with tactical card play under uncertainty. Each round, players must bid on exactly how many tricks they will win, then play cards to achieve that bid. The scoring system rewards accuracy: bidding n and winning exactly n tricks yields $20 + 10n$ points, while missing by k tricks costs $10k$ points. This creates a challenging decision-making environment where players must reason about hidden opponent hands and adapt their strategy based on partial information.

We develop an agent for Wizard using Monte Carlo Tree Search (MCTS) with state sampling to handle hidden information. For card play, we use MCTS with UCB1 selection to balance exploration and exploitation. For bidding, we use forward search with rollouts to evaluate each possible bid. Our key finding is that intelligent bidding (not card play) is the dominant factor in performance. An agent using online planning for bidding achieves a 95% win rate against heuristic opponents, while MCTS applied only to card play provides more modest benefits. We also explore belief-weighted sampling inspired by particle filtering, finding that uniform sampling performs comparably well.

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Figure 1: The Wizard card game, featuring a 60-card deck with special Wizard and Jester cards that always win and lose tricks, respectively.

Related Work

Monte Carlo Tree Search. MCTS builds a search tree by simulating games and using the UCB1 formula to balance exploration and exploitation (Browne et al. 2012). It achieved success in perfect-information games like Go (Silver et al. 2016).

Imperfect Information Games. Card games are harder because players cannot see opponent hands. A common approach is to sample possible opponent hands and run search on each sample (Cowling, Powley, and Whitehouse 2012). This has been used in Bridge (Ginsberg 2001), Skat (Buro et al. 2009), and Hearts (Sturtevant 2008).

Belief State Planning. In POMDPs, tracking exact beliefs over hidden states is often too expensive. Particle filtering approximates the belief using weighted samples (Thrun 2002). We use this approach: weighting sampled opponent hands by how well they explain observed bids.

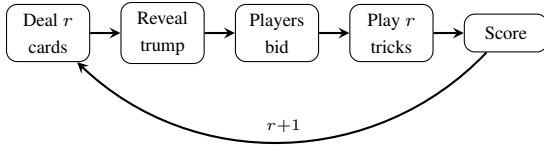


Figure 2: Game flow for one round of Wizard. In round r , each player receives r cards, bids on tricks they expect to win, plays r tricks, then scores based on bid accuracy.

Problem Formulation

The Game of Wizard

Wizard uses a 60-card deck: a standard 52-card deck plus 4 Wizards (always win) and 4 Jesters (always lose). The game consists of multiple rounds, with round r dealing r cards to each player. A trump card is revealed, and players bid sequentially on how many tricks they expect to win.

During play, players must follow the led suit if possible. Wizards always win the trick (first Wizard played if multiple), Jesters always lose, and otherwise the highest trump or highest card of the led suit wins. The scoring function is:

$$R(b, w) = \begin{cases} 20 + 10b & \text{if } w = b \text{ (hit bid exactly)} \\ -10|w - b| & \text{if } w \neq b \text{ (missed by } |w - b| \text{ tricks)} \end{cases}$$

where b is the bid and w is tricks won. This asymmetric reward structure makes accurate bidding critical, because missing a bid by even one trick results in a penalty rather than a bonus.

Bid	Tricks Won	Score
3	3	$20 + 10(3) = +50$
3	2	$-10 3 - 2 = -10$
3	5	$-10 3 - 5 = -20$
0	0	$20 + 10(0) = +20$

Table 1: Scoring examples. Hitting your bid exactly yields a bonus; missing by any amount results in a penalty proportional to the error.

POMDP Formulation

We model Wizard as a Partially Observable Markov Decision Process (POMDP):

States: The true state includes all player hands, cards played, bids made, tricks won by each player, trump suit, current trick leader, and cumulative scores.

Actions: During bidding, choose $b \in \{0, 1, \dots, n\}$ where n is the round number. During play, select a legal card from hand (must follow suit if possible).

Observations: The agent observes its own hand, the trump card, all bids, and all cards played in tricks. Opponent hands remain hidden.

Transitions: Given the true state and actions, transitions follow deterministic game rules. However, from the agent's perspective, opponent actions are uncertain since they depend on hidden hands.

Rewards: Rewards are received at round end according to the scoring function above.

The key challenge is that opponent hands are hidden, creating a large belief space over possible card distributions. With a 60-card deck and 4 players in round 10, each opponent could hold any of $\binom{50}{10}$ possible hands (far too many to enumerate exactly).

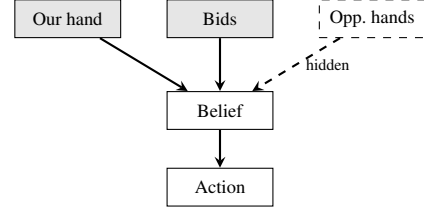


Figure 3: Agent observation structure. Solid boxes are observed; dashed box is hidden. The agent maintains a belief over opponent hands and samples from it to plan actions.

Approach

Our approach uses online belief state planning, drawing on techniques from the course textbook (Kochenderfer, Wheeler, and Wray 2022).

Belief Representation

Since opponent hands are hidden, we must reason about possible card distributions. The space of possible hands is too large for exact enumeration, so we use particle filtering (Section 19.6) to approximate the belief state.

We maintain $N = 100$ particles, where each particle $s^{(i)}$ represents a possible assignment of cards to opponent hands. Particles are weighted by how well they explain observed opponent bids. We model the likelihood that an opponent with hand h would bid b_{obs} using a Gaussian centered on the expected bid $b_{exp}(h)$:

$$w_i = \prod_{j \in \text{opponents}} \exp\left(-\frac{(b_{obs,j} - b_{exp}(h_j^{(i)}))^2}{2\sigma^2}\right)$$

with $\sigma = 1$. This gives higher weight to particles where the sampled hands "explain" the observed bids. The expected bid is calculated using the heuristic baseline described on the following page. When sampling states for planning, we draw from the normalized distribution:

$$P(s^{(i)}) = \frac{w_i}{\sum_{k=1}^N w_k}$$

This focuses computational effort on plausible opponent hands rather than uniform random sampling.

Online Planning for Bidding

For bidding decisions, we use lookahead with rollouts (Section 22.1). For each possible bid $b \in \{0, 1, \dots, n\}$:

1. Sample $m = 50$ states from the belief distribution

2. For each sampled state, simulate the entire round to completion (all players use heuristic card play)
3. Compute the score we would receive given bid b and the number of tricks won

We select the bid with highest average score: $b^* = \arg \max_b \mathbb{E}[\text{score} | \text{bid} = b]$. Unlike MCTS, this is flat Monte Carlo evaluation (we simply average outcomes across sampled states without building a search tree).

MCTS for Card Play

For card play, we use Monte Carlo Tree Search (MCTS, Section 22.5). Each iteration:

1. **Sample:** Draw a state from the belief distribution.
2. **Select:** Choose an action using UCB1 (Auer, Cesa-Bianchi, and Fischer 2002):

$$UCB1(a) = \bar{Q}(a) + C \sqrt{\frac{\ln N_{parent}}{N(a)}}$$

where $\bar{Q}(a)$ is the average reward, $N(a)$ is the visit count, and $C = \sqrt{2}$ balances exploration vs exploitation.

3. **Simulate:** From the selected action, play out the remainder of the round (all players use heuristic play) and compute the resulting score.

4. **Backpropagate:** Update statistics: $N(a) \leftarrow N(a) + 1$ and $Q_{total}(a) \leftarrow Q_{total}(a) + R$.

After $m = 100$ iterations, we select the action with the highest visit count. By aggregating statistics across many sampled states, we approximate the true Q-values despite hidden information.

Algorithm 1 MCTS with State Sampling for Card Play

Require: Current hand h , belief B , legal actions A , iterations m

- 1: Initialize $N(a) \leftarrow 0, Q(a) \leftarrow 0$ for all $a \in A$
 - 2: **for** $i = 1$ to m **do**
 - 3: $s \sim B$ {Sample state from belief}
 - 4: **if** $\exists a \in A$ with $N(a) = 0$ **then**
 - 5: $a^* \leftarrow$ unexplored action
 - 6: **else**
 - 7: $a^* \leftarrow \arg \max_a \frac{Q(a)}{N(a)} + C \sqrt{\frac{\ln \sum_a N(a)}{N(a)}}$
 - 8: **end if**
 - 9: $R \leftarrow \text{ROLLOUT}(s, a^*)$ {Simulate to round end}
 - 10: $N(a^*) \leftarrow N(a^*) + 1$
 - 11: $Q(a^*) \leftarrow Q(a^*) + R$
 - 12: **end for**
 - 13: **return** $\arg \max_a N(a)$
-

Belief Updates During Play

We refine beliefs as opponents play cards:

1. If an opponent plays a card, they must have had it (eliminate inconsistent particles)
2. If an opponent doesn't follow suit, they lack that suit (strong constraint)

Particles are resampled based on the effective sample size (ESS) of the current belief, defined as:

$$N_{ESS} = \frac{1}{\sum_{i=1}^N w_i^2}$$

When the effective sample size of the belief is below some threshold, N_{thresh} , we resample new particles from the current belief. This prevents the probability mass of the belief from accumulating over a very small number of particles, which can result in weight degeneracy (i.e. only a very small number of particles are actually useful, drastically reducing the effective sample size).

The re-sampling favors particles with higher weights, and also eliminates the possibility of propagating any particles inconsistent with our observations since they will have assigned weights of zero.

We empirically determined that $N_{thresh} = N/2$ was an adequate threshold value for resampling.

Heuristic Baselines

Heuristic bidding counts Wizards (+1), high trump (+0.6), and off-suit Aces (+0.3), then applies a conservative multiplier. Heuristic card play uses high cards when trying to win, low cards when trying to lose.

Implementation

Game Simulator

We implemented a complete Wizard game simulator in Python supporting 3-6 players. The simulator handles all game rules: the 60-card deck (52 standard cards plus 4 Wizards and 4 Jesters), trump determination, suit-following requirements, and trick resolution. A full game consists of $\lfloor 60/n \rfloor$ rounds for n players (15 rounds for 4 players). Each game takes approximately 30 seconds with our MCTS agent (100 simulations per card play decision).

Hyperparameters

Table 2 lists the hyperparameters used in our experiments. These were chosen based on preliminary experiments balancing computation time against decision quality.

Parameter	Symbol	Value
MCTS iterations (card play)	m	100
Bid simulations	–	50
Number of particles	N	100
ESS resample threshold	N_{thresh}	$N/2$
UCB exploration constant	C	$\sqrt{2}$
Belief weighting σ	σ	1.0
Number of players	n	4

Table 2: Hyperparameters for our MCTS agent.

Experimental Setup

All experiments were run with 4 players over 100 games each. In the ablation study, one agent uses full online planning, one uses online card play with heuristic bidding, and

two use pure heuristics. For the belief comparison, one agent uses particle filtering, one uses uniform sampling, and two use heuristics. Win rates are computed as the fraction of games where each agent achieved the highest final score. Average scores are computed across all 15 rounds of each game.

Experiments

Ablation Study: Bidding vs Card Play

Agent	Win Rate	Avg Score
Online planning (bid + play)	95%	+207.8
Online play + heuristic bid	4%	-29.6
Heuristic (baseline)	1%	-143.8

Table 3: Ablation study (100 games). Online planning for bidding provides +237.4 point improvement over heuristic bidding.

Key finding: Online planning for bidding is the dominant factor. The full agent achieves 95% win rate, while MCTS card play with heuristic bidding wins only 4% of games.

MCTS Card Play vs Heuristic

To isolate the impact of MCTS card play, we ran a separate experiment where all agents use heuristic bidding:

Agent	Win Rate	Avg Score
MCTS card play	53%	-63.2
Heuristic (baseline)	16% each	-151.1

Table 4: MCTS card play vs heuristic (100 games). MCTS card play provides +87.9 point improvement.

MCTS card play does provide benefit (+87.9 points, 53% win rate vs expected 25%), but this improvement is not as important as the +237.4 point gain from online bidding.

Particle Filtering vs Uniform Sampling

Agent	Win Rate	Avg Score
Uniform sampling	54%	+154.1
Particle filtering	46%	+112.6
Heuristic (baseline)	0%	-103.9

Table 5: Particle filtering vs uniform sampling (100 games). Uniform sampling slightly outperforms particle filtering.

Contrary to our expectations, uniform sampling slightly outperformed particle filtering (54% vs 46% win rate, +154.1 vs +112.6 average score). Both methods decisively beat heuristic opponents (0% win rate for heuristics).

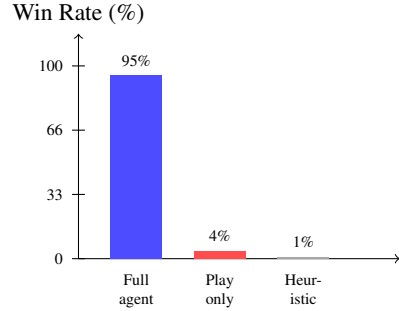


Figure 4: Win rates from ablation study. Online bidding is the key factor: the full agent (with online bidding) wins 95% of games, while MCTS card play alone wins only 4%.

Discussion

Why Bidding Matters More Than Card Play

Our ablation study reveals a striking finding: online planning for bidding provides a +237 point improvement (from -29.6 to +207.8), while MCTS card play alone provides only +88 points. The full agent wins 95% of games; without online bidding, it wins only 4%.

In Wizard, the bid you choose determines your entire round strategy. A bad bid (too high or too low) puts you in an unwinnable situation. No amount of clever card play can recover from bidding 5 when you can only win 2 tricks. Conversely, a good bid aligned with your hand strength makes the round easier regardless of card play quality. This aligns with the game’s own marketing: the Wizard box declares “The winning is in the BIDDING!”

MCTS card play does help (53% win rate vs 25% expected, +88 points), but this benefit is smaller compared to online bidding. The heuristic card play captures most strategic value by playing high cards when needing tricks and low cards otherwise. The heuristic bidding, however, misses important interactions like opponents also holding strong cards.

Why Particle Filtering Underperformed

Surprisingly, our particle filtering approach slightly underperformed uniform sampling (46% vs 54% win rate). We hypothesize several reasons:

Noisy bid signals: Opponent bids may not reliably indicate hand strength. A bid of 2 could come from two guaranteed Wizards or from several medium-strength cards. Our Gaussian likelihood model may be too simplistic.

Sufficient samples: With 50 samples, uniform sampling already explores the space well. The marginal benefit of focusing on “likely” hands may not outweigh the cost of missing unlikely but important scenarios.

Heuristic bid model mismatch: We weight particles by how well hands explain bids assuming heuristic bidding. But our MCTS opponents also use online bidding, creating a mismatch.

Limitations

Our approach has several limitations. First, we assume opponents follow heuristic play during rollouts, which may not match their actual strategies. Second, our observation model for particle weighting uses a simple Gaussian based on expected bids, which may not capture all bidding behaviors. Third, we don't model individual opponent tendencies or learn from game history.

Future Work

Future improvements could include: (1) learning opponent-specific bid models from game history, and (2) more sophisticated particle weighting that accounts for opponent skill level.

Conclusion

We developed an agent for the card game Wizard using Monte Carlo Tree Search (MCTS) with state sampling and UCB1 selection for card play, combined with forward search rollouts for bid evaluation. Our key finding is that intelligent bidding is far more important than card play optimization. Online planning for bidding achieves 95% win rate and +237 points improvement, while MCTS card play alone provides only +88 points. We also implemented particle filtering to weight opponent hand samples by bid consistency, but found that uniform sampling slightly outperformed it (54% vs 46% win rate), suggesting that simple approaches work well when sufficient samples are used. These results suggest that in Wizard, and perhaps other trick-taking games with bidding, the commitment decision (what to bid) may matter more than the tactical execution (how to play cards).

Contributions

Joana Mizrahi: Game simulator implementation, Heuristic agent implementation, MCTS agent development, experiment design and execution, paper writing.

Jonathan Dumanski: MCTS bidding evaluation, particle filtering belief representation, experimental analysis, paper writing.

References

- Auer, P.; Cesa-Bianchi, N.; and Fischer, P. 2002. Finite-time analysis of the multiarmed bandit problem. *Machine Learning* 47(2):235–256.
- Browne, C. B.; Powley, E.; Whitehouse, D.; Lucas, S. M.; Cowling, P. I.; Rohlfshagen, P.; Tavener, S.; Perez, D.; Samothrakis, S.; and Colton, S. 2012. A survey of Monte Carlo tree search methods. *IEEE Transactions on Computational Intelligence and AI in Games* 4(1):1–43.
- Buro, M.; Long, J. R.; Furtak, T.; and Sturtevant, N. R. 2009. Improving state evaluation, inference, and search in trick-taking card games. In *Proceedings of the Twenty-First International Joint Conference on Artificial Intelligence*, 1407–1413.
- Cowling, P. I.; Powley, E. J.; and Whitehouse, D. 2012. Information set Monte Carlo tree search. In *IEEE Transactions*

on Computational Intelligence and AI in Games, volume 4, 120–143. IEEE.

Ginsberg, M. L. 2001. GIB: Imperfect information in a computationally challenging game. In *Journal of Artificial Intelligence Research*, volume 14, 303–358.

Kochenderfer, M. J.; Wheeler, T. A.; and Wray, K. H. 2022. *Algorithms for Decision Making*. MIT Press.

Silver, D.; Huang, A.; Maddison, C. J.; Guez, A.; Sifre, L.; Van Den Driessche, G.; Schrittwieser, J.; Antonoglou, I.; Panneershelvam, V.; Lanctot, M.; et al. 2016. Mastering the game of Go with deep neural networks and tree search. *Nature* 529(7587):484–489.

Sturtevant, N. R. 2008. An analysis of UCT in multi-player games. *ICGA Journal* 31(4):195–208.

Thrun, S. 2002. Particle filters in robotics. *Proceedings of the Conference on Uncertainty in Artificial Intelligence* 511–518.